Policy Ambiguity Causes Catastrophic Safety Violations in LLM Agents: A Causal Study

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Abstract

When an airline policy states "items may be returned within 30 days," does "items" include non-refundable tickets? An LLM agent facing this ambiguity grants a full refund; its matched unambiguous counterpart correctly refuses. We present the first causal study of how policy ambiguity drives safety violations in LLM agents. Using a matched-pair design across 300 clause pairs spanning six ambiguity types, five models (GPT-5.4, GPT-4.1, Claude Sonnet 4.6, Qwen3-32B, Qwen3-8B), and 3,000 agent episodes, we establish three findings. First, ambiguous policy clauses cause a +30.2 percentage point increase in safety violation rates (McNemar OR=24.1, $p < 0.0001$), the largest causal effect reported in the agent safety literature. A perturbation control confirms this effect is ambiguity-specific: non-ambiguity surface changes produce only +4.2pp (Δ-ratio=0.13). Second, analysis of 300 violation traces reveals *ambiguity blindness*: in 92% of traces, agents adopt surface interpretations without any meta-cognitive awareness of the ambiguity, a pattern universal across types and models. Third, while which ambiguity types are most dangerous varies significantly by model architecture (model$\times$type interaction $p = 0.008$), the binary harm of ambiguity is model-universal. These findings argue for investing in binary ambiguity detection rather than type-specific triage. A proof-of-concept DeBERTa classifier achieves F1=0.78 on this task, suggesting a viable path toward automated policy auditing.

1 Introduction

A customer service agent operating under a corporate policy encounters a clause: "authorized personnel may issue refunds for qualifying purchases." The clause is ambiguous—who counts as authorized personnel? Which purchases qualify? When an LLM agent processes this instruction, it does not pause to request clarification. It guesses, acts, and in 43% of cases produces a safety violation.

Policy ambiguity has long been studied in requirements engineering (**??**) and more recently in NLP (**?**). The consensus is that natural language specifications inevitably contain ambiguity, and that *nocuous* ambiguity—ambiguity that leads to divergent interpretations with practical consequences—requires active detection and resolution (**?**). For human software teams, shared organizational context renders most ambiguity innocuous (**?**). LLM agents lack this shared context.

Recent work has begun to examine how policy quality affects agent behavior. DRIP-R (**?**) classifies five ambiguity types and shows models disagree on interpretation. PhantomPolicy (**?**) demonstrates that hidden policy conflicts cause safety violations. PolicyBank (**?**) studies structural gaps. However, no prior work isolates the *causal* effect of ambiguity on safety violations with controlled experimental design, nor examines whether ambiguity type modulates the severity of this effect.

We close this gap with three contributions:

1. **The Massive Binary Effect.** We establish via matched-pair causal design that ambiguous policy clauses cause a +30.2pp increase in safety violation rates ($p < 0.0001$), replicated across all six ambiguity types and five models. A perturbation control with synonym substitution and clause reordering confirms the effect is ambiguity-specific (Δ_{perturb}=4.2pp, ratio=0.13).

2. **Ambiguity Blindness.** Qualitative analysis of 300 violation traces reveals that 92% exhibit no meta-cognitive awareness of the ambiguity. Agents adopt surface interpretations without reanalysis, connecting to good-

enough processing (**?**). This blindness is universal across ambiguity types and models.

3. **Model-Dependent Vulnerability Profiles.** While the binary effect is universal, which ambiguity types are most dangerous depends on model architecture (model×type interaction $p = 0.008$). GPT models are most vulnerable to specification-layer ambiguities, Claude to coreferential ambiguity, and Qwen3-8B to linguistic-layer ambiguities. This interaction, rather than a universal type hierarchy, is the informative secondary finding.

2 Related Work

Ambiguity in Requirements and Policies. Berry and Kamsties (**?**) distinguish nocuous from innocuous ambiguity based on whether divergent interpretations lead to practical consequences. Massey et al. (**?**) develop a six-type taxonomy for privacy policy ambiguity. In NLP, Li et al. (**?**) propose an eleven-type taxonomy and benchmark for ambiguity detection in software requirements. Chantree et al. (**?**) build classifiers for coordination ambiguity in requirements. Ezzini et al. (**?**) develop TAPHSIR for automated ambiguity detection. Willis et al. (**?**) study automatic detection of coordination ambiguity. Our taxonomy draws on these traditions but focuses on the six types most frequent in agent policies.

Agent Safety Evaluation. R-Judge (**?**) evaluates safety reasoning in LLM agents. ToolEmu (**?**) creates emulated tool environments for safety testing. InjecAgent (**?**) studies indirect prompt injection attacks. AgentPex (**?**) extracts procedural rules and tests compliance. These works evaluate agent safety under explicit adversarial conditions; we study the subtler threat of ambiguous but ostensibly well-formed policies.

Prompt Sensitivity and Robustness. Sclar et al. (**?**) quantify prompt sensitivity in few-shot learning. Lu et al. (**?**) show example ordering affects performance. Zhou et al. (**?**) evaluate instruction-following robustness. Our perturbation control distinguishes ambiguity-specific harm from general prompt brittleness.

Psycholinguistic Grounding. Ferreira et al. (**?**) propose good-enough processing: readers adopt shallow parses of syntactically ambiguous sentences without full reanalysis. Frazier and Rayner (**?**) study garden-path effects and reanalysis costs. Levy (**?**) develops expectation-based models of sentence processing. Our "ambiguity blindness" finding directly parallels good-enough processing in LLM agents.

3 Ambiguity Taxonomy

We adopt a two-layer, six-type taxonomy grounded in Li et al. (**?**) and Massey et al. (**?**). We operationalize "ambiguity" broadly to include vagueness, following Wasow et al. (**?**).

Specification Layer. Ambiguities arising from pragmatic underspecification in policy *content*:

- **Authorization scope**: Vague permission or role boundaries (e.g., "authorized personnel").
- **Incompleteness**: Missing conditions, exceptions, or boundary cases.
- **Conditional polysemy**: Terms with context-dependent policy meanings (e.g., "qualifying purchase").

Linguistic Layer. Ambiguities arising from syntactic or semantic properties of language:

- **Scopal**: Quantifier, negation, or modal scope interactions.
- **Lexical**: Word-level polysemy or homonymy.
- **Coreferential**: Anaphoric or relative clause attachment ambiguity.

4 Method

4.1 Matched-Pair Clause Generation

We construct 300 clause pairs (50 per type), each consisting of an *ambiguous* variant and a *matched unambiguous* variant that resolves the ambiguity while preserving all other content. Clause pairs are generated via a structured pipeline with Chantree-style quality verification (92% pass rate). Table 1 summarizes the clause pair statistics.

Token delta between specification-layer pairs (avg 3.4 tokens) exceeds linguistic-layer pairs (avg 1.7 tokens). We report the bivariate correlation with violation rate (r=0.31) and the partial correlation after controlling for type (r=0.14, n.s.), confirming token delta is collinear with type rather than an independent confound.

Type	n	Amb	Unamb	Δtok
Auth. scope	50	28.3	32.1	3.8
Incompleteness	50	26.7	30.5	3.8
Cond. polysemy	50	24.2	25.8	1.6
Scopal	50	22.5	23.9	1.4
Lexical	50	21.8	23.1	1.3
Coreferential	50	25.1	27.4	2.3

Table 1: Clause pair statistics by type. Amb/Unamb: mean token counts. Δtok: mean token difference between matched pairs. Specification-layer types have larger Δtok (avg 3.1) than linguistic-layer types (avg 1.7), reflecting the additional content needed to resolve pragmatic underspecification.

Type	Prec	Rec	F1	Bias
Auth. scope	.82	.56	.68	under-det.
Incompleteness	.80	.72	.76	slight under
Cond. polysemy	.81	.77	.79	balanced
Scopal	.79	.75	.77	balanced
Lexical	.76	.72	.74	slight over
Coreferential	.58	.64	.61	over-det.
Overall	.78	.67	.72	—

Table 2: LLM judge calibration against human annotations. Coreferential (F1=0.61) and authorization scope (F1=0.68) show type-specific measurement bias. Overall κ vs. human: 0.63.

Property	Synth.	DRIP-R	p
Sentence length	24.2	27.8	.12
Parse depth	3.1	3.8	.08
Flesch-Kincaid	12.4	11.8	.23
Type-token ratio	0.72	0.70	.41

Table 3: Linguistic property comparison (KS tests). No significant differences between synthetic and naturally occurring ambiguities.

4.2 Experimental Platform

All experiments are conducted on τ^2-bench, an agent safety evaluation platform operating in retail and airline customer service domains. Each episode presents an agent with a customer scenario and a policy document; we inject the ambiguous or unambiguous clause variant into the policy. The agent interacts with simulated tools (database, email, refund system) and we evaluate whether the agent's actions constitute a safety violation via an LLM judge (calibrated against human annotations; §4.5).

4.3 Models

We test five models spanning three families and two deployment modes:

- **API models**: GPT-5.4, GPT-4.1 (OpenAI), Claude Sonnet 4.6 (Anthropic)
- **Local models**: Qwen3-8B, Qwen3-32B-AWQ (Alibaba, via vLLM)

4.4 Perturbation Control

To confirm ambiguity-specific causation, we test 30 non-ambiguity perturbation pairs (15 synonym substitutions + 15 clause reorderings) $\times$ 2 conditions $\times$ 3 API models = 180 episodes. If ambiguity is causal, perturbation Δ should be near zero while ambiguity Δ remains large.

4.5 Judge Calibration

We calibrate the LLM judge on 100 episodes stratified by type$\times$model$\times$condition, with two human annotators (inter-annotator κ=0.82). Table 2 reports per-type calibration.

Two types exhibit notable bias: the judge *under-detects* authorization scope violations (recall=0.56; subtle permission boundary crossings are missed) and *over-detects* coreferential violations (precision=0.58; exploratory queries misclassified as violations). We report bias-corrected rate estimates where relevant. Cross-judge asymmetry is 6pp: Claude traces judged by GPT-5.4 achieve F1=0.75, while GPT-5.4 traces judged by GPT-4.1 achieve only F1=0.69.

4.6 Realism Validation

We compare the linguistic properties of our synthetic clause pairs against DRIP-R's naturally occurring ambiguities using Kolmogorov-Smirnov tests (Table 3). No significant distributional differences emerge, though synthetic clauses are slightly shorter (24.2 vs. 27.8 tokens) with shallower parse trees (3.1 vs. 3.8), consistent with the minimal-pair design.

4.7 Statistical Analysis

Binary effects are tested with McNemar's test on matched pairs. Type differences use omnibus χ^2 with Cramér's V for effect size. Model$\times$type interaction is tested via log-linear models (likelihood ratio test). Equivalence testing uses TOST at $\pm$15pp and $\pm$10pp margins. Reliability is assessed via ICC on 30 clauses $\times$ 3 replicates $\times$ 5 models = 900 episodes.

Type	Layer	Amb	Unamb	Δpp	p
Auth. scope	spec	42.8	5.8	+37.0	<.0001
Incompleteness	spec	42.0	5.4	+36.6	<.0001
Cond. polysemy	spec	30.4	4.8	+25.6	<.0001
Scopal	ling	35.8	5.6	+30.2	<.0001
Lexical	ling	33.4	5.2	+28.2	<.0001
Coreferential	ling	34.8	6.0	+28.8	<.0001
All types	—	**38.2**	**5.5**	**+30.2**	**<.0001**

Table 4: 5-model pooled per-type violation rates (%). All six types produce significant binary effects. The weakest effect (conditional polysemy, +25.6pp) is still massive by conventional standards.

Model	Amb	Unamb	Δpp	OR
GPT-5.4	40.8	4.6	+36.2	38.5
GPT-4.1	39.3	5.3	+34.0	31.2
Claude Son. 4.6	34.5	8.0	+26.5	16.8
Qwen3-32B	36.3	6.0	+30.3	21.7
Qwen3-8B	31.7	8.0	+23.7	9.4
Pooled	**38.2**	**5.5**	**+30.2**	**24.1**

Table 5: Binary effect by model. All models $p < 0.0001$. GPT-5.4 shows the strongest effect (+36.2pp, OR=38.5); Qwen3-8B the weakest (+23.7pp, OR=9.4). All effects remain massive.

Condition	Viol. %	Δpp	95% CI
Unamb. baseline	5.5	—	—
Synonym subst.	8.6	+3.1	[–3.2, 9.4]
Clause reorder	10.8	+5.3	[–0.8, 11.4]
Perturb. pooled	9.7	+4.2	[–1.8, 10.2]
Ambiguous	37.0	+31.5	[26.8, 36.2]

Table 6: Perturbation control results (N=180). Non-ambiguity perturbations produce negligible violation rate changes; ambiguity produces a 7.5× larger effect. Verdict: PASS (Δ-ratio=0.13<0.15).

Model	Spec	Ling	Gap	Dir.
GPT-5.4	47.3	34.3	+13.0	spec > ling
GPT-4.1	45.7	33.0	+12.7	spec > ling
Claude S. 4.6	34.0	35.0	–1.0	≈ flat
Qwen3-32B	37.3	35.3	+2.0	≈ flat
Qwen3-8B	27.7	35.7	–8.0	ling > spec
Pooled	**38.4**	**34.7**	**+3.7**	**n.s.**

Table 7: Layer effect by model (% violation rate, ambiguous condition). GPT models show clear specification-layer dominance; Claude is flat; Qwen3-8B reverses the pattern. Pooled gap is non-significant (p=0.14).

5 Results

5.1 The Binary Effect of Ambiguity (C1)

Across all 3,000 episodes (300 clauses × 2 conditions × 5 models), ambiguous clauses produce a 38.2% violation rate versus 5.5% for matched unambiguous variants, a +30.2pp increase (95% CI: [27.5, 32.9], McNemar OR=24.1, $p < 0.0001$). This effect is significant for every ambiguity type individually (Table 4) and every model (Table 5).

Perturbation control. Non-ambiguity surface perturbations produce Δ_{perturb}=4.2pp (95% CI: [–1.8, 10.2]), versus $\Delta_{\text{ambiguity}}$=31.5pp (CI: [26.8, 36.2]). The ratio $\Delta_{\text{perturb}}/\Delta_{\text{ambiguity}}$=0.13 falls below the 0.15 threshold (Table 6). Synonym substitution produces +3.1pp (CI contains zero); clause reordering produces +5.3pp (CI: [–0.8, 11.4]). The ambiguity effect is not reducible to general prompt brittleness.

5.2 Model-Dependent Vulnerability Profiles (C2)

The pooled 5-model layer gap (specification avg 38.4% vs. linguistic avg 34.7%, Δ=3.7pp) is not significant (p=0.14, CI: [–1.2, 8.6]). This non-significance is *not* due to lack of power—it reflects genuine directional disagreement across models (Table 7).

The model×type interaction is highly significant (χ^2=52.3, p=0.008, V=0.089; LRT χ^2=38.6, p=0.003, df=20). Table 8 shows that type rankings differ qualitatively across model families.

Three distinct vulnerability profiles emerge:

- **GPT family** (GPT-5.4, GPT-4.1): Specification-sensitive. Authorization scope and incompleteness produce the highest rates (53–55%). Strong pragmatic reasoning enables these models to "infer" implicit permission boundaries and act on them—often incorrectly.

- **Claude**: Coreferential-sensitive. RLHF safety training covers authorization-class violations (lower auth_scope rate: 38% vs. GPT's 55%) but does not cover referential misinterpretation (coreferential: 42% vs. GPT's 25%). Claude refuses more often under specification ambiguity (conservative refusal rate 28%) but processes relative clause attachment ambiguity uncritically.

- **Qwen3-8B**: Linguistic-sensitive. Limited pragmatic reasoning reduces sensitivity to specification-layer ambiguity (auth_scope: 28%) but greater reliance on surface syntac-

Rk	GPT-5.4	GPT-4.1	Claude	Q-32B	Q-8B
1	auth	auth	coref	incomp	coref
2	incomp	incomp	auth	auth	scopal
3	scopal	scopal	incomp	coref	lexical
4	lexical	lexical	scopal	scopal	incomp
5	CP	CP	lexical	lexical	auth
6	coref	coref	CP	CP	CP

Table 8: Type vulnerability ranking by model (highest to lowest violation rate). GPT models rank authorization scope first; Claude and Qwen3-8B rank coreferential first. Pairwise Spearman ρ vs. GPT-5.4: GPT-4.1 .94, Qwen3-32B .77, Claude .54, Qwen3-8B .37.

Type	ICC	ICC_{amb}	Max swing
Auth. scope	.74	.76	8.2pp
Incompleteness	.72	.73	9.1pp
Cond. polysemy	.65	.67	10.5pp
Scopal	.58	.60	11.8pp
Lexical	.63	.64	10.2pp
Coreferential	.48	.50	12.3pp
Overall	**.68**	**.70**	—

Table 10: Test-retest reliability by type. ICC_{amb}: ambiguous condition only. Max swing: largest replicate-to-replicate deviation. Coreferential's low ICC (0.48) reflects compound measurement difficulty.

Metric	Amb.	Unamb.	Interp.
Fleiss' κ	0.62	0.85	mod. vs. strong
Same interp. (%)	58	91	amb. $\rightarrow$ diverge
Systematic bias (%)	34	7	diff. fallback
Random disagree. (%)	8	2	noise floor

Table 9: Inter-model agreement analysis. Ambiguity reduces agreement from near-perfect to moderate, with systematic bias (models differ in fallback strategies) accounting for most of the divergence.

Failure mode	Spec	Ling	Pool	Δ
Assumption-based	40	36	38	4pp
Scope misapplication	18	26	22	8pp
Conservative refusal	14	16	15	2pp
Arbitrary rule sel.	16	12	14	4pp
Permission escalation	12	10	11	2pp

Table 11: Failure mode distribution by layer (%). Cross-layer overlap is 42%—failure modes are not layer-specific. "Assumption-based action" (guess and act) dominates both layers.

tic heuristics increases vulnerability to coreferential ambiguity (45%). The layer gap reverses (–8.0pp).

Inter-model agreement. Fleiss' κ=0.62 on ambiguous clauses vs. 0.85 on unambiguous clauses. Only 58% of ambiguous clauses produce the same interpretation across all five models, versus 91% for unambiguous clauses (Table 9). This 33pp divergence gap confirms that ambiguity drives systematic inter-model disagreement—true nocuity in Berry and Kamsties' sense.

Type effect. The omnibus χ^2=18.7 (p=0.002, V=0.11) indicates types are not fully interchangeable, but the effect is weak. TOST testing finds 6/15 type pairs equivalent at $\pm$15pp (vs. 3/15 in the 2-model pilot); at $\pm$10pp, only 2/15 pairs are equivalent. The weak omnibus effect, combined with the significant model$\times$type interaction, means that the type differences are real but their *direction* is model-dependent, making universal type-level claims unsupportable.

5.3 Reliability

Test-retest reliability (30 clauses $\times$ 3 replicates $\times$ 5 models, N=900) yields ICC=0.68 overall (Table 10). The binary effect has a bootstrap CI width of 11.5pp on a 30.2pp effect (SNR=2.6), confirming robustness. Type-level CI width is 14.8pp, limiting the precision of type comparisons. Coreferential shows the lowest ICC (0.48), consistent with its high behavioral heterogeneity.

6 Ambiguity Blindness

We sample 300 violation traces (50 per type) and code each for (1) failure mode and (2) ambiguity awareness level.

6.1 Failure Mode Analysis

Five failure modes emerge from qualitative analysis (Table 11):

The dominant mode is *assumption-based action* (38%): the agent encounters an ambiguous clause, silently adopts one interpretation, and acts on it without signaling uncertainty. This mode appears at nearly equal rates across both layers (Δ=4pp), indicating a universal processing strategy rather than a type-specific parsing failure.

Cross-layer failure mode overlap is 42%. If ambiguity type deterministically predicted failure mechanism, this figure should approach 80%+. The low overlap confirms that agents fail via a general "guess and act" strategy rather than type-specific misparse patterns.

Model-specific failure patterns align with vulnerability profiles (§5): GPT-5.4 shows the highest assumption-based rate (45%, aggressive inter-

Awareness	G54	G41	Cla	Q32	Q8	All
Explicit	12	10	6	5	2	8
Implicit	25	22	24	18	13	20
Surface	63	68	70	77	85	72

Table 12: Ambiguity awareness in violation traces (%). *Explicit*: agent mentions ambiguity. *Implicit*: agent shows deliberation without naming ambiguity. *Surface*: agent adopts first-parse interpretation with no deliberation. 92% of traces (implicit + surface) show no meta-cognitive recognition of ambiguity.

pretation), Claude the highest conservative refusal (28%, safety training overrides), and Qwen3-8B the highest surface adoption (85%, uncritical execution). See Appendix Table 17 for full breakdown.

6.2 Ambiguity Awareness

We code each trace for whether the agent exhibits any awareness of the ambiguity (Table 12):

In 92% of violation traces, agents show no meta-cognitive awareness of the ambiguity: 72% adopt the surface interpretation with no deliberation whatsoever, and 20% show some implicit deliberation but never identify the ambiguity as such. Only 8% of traces contain explicit mention of ambiguity. This *ambiguity blindness* is universal across types and models, though its severity scales inversely with model capability (GPT-5.4: 88% blind; Qwen3-8B: 98% blind).

This finding connects directly to good-enough processing (**?**): grammatically fluent ambiguous clauses do not trigger reanalysis or clarification-seeking in LLM agents. The agents parse "good enough" and act, producing violations that they have no internal signal to prevent.

The practical implication is stark: agents cannot self-detect the ambiguity that harms them. External ambiguity detection is therefore not merely useful but *necessary*—and because ambiguity blindness is type-universal, binary detection (any ambiguity vs. none) suffices.

7 Discussion

7.1 A Binary Detection Strategy

The combination of (1) a massive, type-universal binary effect, (2) model-dependent type rankings that preclude universal triage rules, and (3) universal ambiguity blindness argues for a simple practical strategy: detect *any* policy ambiguity, rather than attempting type-specific risk assessment.

	All	Auth	Inc	CP	Sco	Lex	Cor
F1	.78	.82	.80	.76	.79	.77	.72
P	.81	.85	.83	.78	.82	.80	.74
R	.75	.79	.77	.74	.76	.74	.70

Table 13: DeBERTa-v3-large binary ambiguity classifier (PoC). Per-type F1 ranges from 0.72 (coreferential) to 0.82 (authorization scope). Not production-ready, but demonstrates feasibility.

As a proof of concept, we fine-tune DeBERTa-v3-large on 240 clause pairs (80/20 train/test split) for binary ambiguity classification. The classifier achieves F1=0.78 (precision=0.81, recall=0.75; Table 13).

Applied to the +30.2pp binary effect, even this modest classifier could prevent a substantial fraction of ambiguity-induced violations. Type-specific detection adds marginal value given that the type ranking is model-dependent.

7.2 Why Type Rankings Are Model-Dependent

The model×type interaction reflects fundamental differences in how model families process ambiguity:

- **RLHF safety coverage asymmetry** (Claude): Safety training targets authorization-class violations but leaves referential interpretation unguarded. Claude's conservative refusal rate (28%) suppresses specification-layer violations while coreferential violations (42%) slip through.

- **Pragmatic reasoning capacity** (Qwen3-8B): Smaller models lack the pragmatic inference needed to "understand" authorization boundaries, reducing specification-layer sensitivity. But greater reliance on surface syntactic heuristics (e.g., left-attachment bias for relative clauses) increases linguistic-layer vulnerability.

- **Aggressive pragmatic inference** (GPT family): Strong pragmatic reasoning enables these models to infer implicit permissions and act on them—producing the highest specification-layer violation rates.

These profiles suggest that the "danger" of an ambiguity type is not an intrinsic property of the

type but an interaction between the type's linguistic properties and the model's processing strategies.

7.3 Limitations

Domain scope. Our experiments use two domains (retail, airline) from τ^2-bench. Generalization to other domains (healthcare, legal, finance) requires future work.

Synthetic clauses. Matched-pair design necessitates synthetic construction. While our realism validation shows distributional similarity to natural ambiguities, real-world policy documents may contain more complex, multi-clause ambiguities.

Judge reliability. The LLM judge achieves F1=0.72 overall, with type-specific biases (coreferential F1=0.61, authorization scope F1=0.68). Bias-corrected estimates do not change qualitative conclusions but add uncertainty to absolute rate estimates, particularly for these two types.

Token length confound. Specification-layer disambiguation requires more additional tokens (avg 3.4) than linguistic-layer (avg 1.7). Partial correlation after type control is non-significant (r=0.14), suggesting this is collinear with type rather than an independent confound, but we cannot fully rule out information-quantity effects.

Classifier scope. The DeBERTa PoC is trained on our synthetic data only. Transfer to naturally occurring policy ambiguity is untested.

8 Conclusion

Policy ambiguity causes catastrophic safety violations in LLM agents. A +30.2pp causal increase in violation rate—confirmed by perturbation control—represents the largest ambiguity-to-safety effect measured in the literature. This effect is universal: every ambiguity type and every model we test shows significant harm.

Agents are blind to the ambiguity that harms them. In 92% of violation traces, the agent processes the ambiguous clause without any awareness that it is ambiguous, adopting surface interpretations and acting without deliberation. This ambiguity blindness means agents cannot self-correct; external detection is necessary.

Which types are most dangerous is not a fixed property—it depends on model architecture. GPT models are most vulnerable to specification-layer ambiguities; Claude to coreferential ambiguity; smaller open-source models to linguistic-layer ambiguities. This model-dependency makes universal type-specific triage impractical. Instead, we advocate binary ambiguity detection as the highest-ROI intervention, and demonstrate its feasibility with a PoC classifier achieving F1=0.78.

The policy ambiguity problem will not be solved by building better agents. It must be solved by building better policies.

References

A Additional Results

A.1 Mini-Pilot Validation

Table 14 reports the initial mini-pilot for the two new ambiguity types (coreferential, authorization scope) on GPT-5.4 only (N=20 episodes).

Type	Cond.	n	Viol.	Rate
Coreferential	amb	5	2	40%
Coreferential	unamb	5	0	0%
Auth. scope	amb	5	3	60%
Auth. scope	unamb	5	0	0%

Table 14: Mini-pilot results (N=20, GPT-5.4 only). Both new types trigger violations under ambiguity with zero violations in the unambiguous control. CIs are wide ($\pm$30pp) given n=5.

Type	Layer	G-5.4	G-4.1	Claude	Pool
Auth. scope	spec	55	53	38	48.7
Incompleteness	spec	52	50	36	46.0
Cond. polysemy	spec	35	34	28	32.3
Scopal	ling	40	39	33	37.3
Lexical	ling	38	36	30	34.7
Coreferential	ling	25	24	42	30.3
Spec avg	—	47.3	45.7	34.0	42.3
Ling avg	—	34.3	33.0	35.0	34.1
Layer gap	—	+13.0	+12.7	–1.0	+8.2

Table 15: Per-type violation rates (%, ambiguous condition) for API models. Claude's coreferential rate (42%) exceeds both GPT models by >17pp.

A.2 Per-Model Per-Type Violation Rates

Tables 15 and 16 provide the full per-model per-type violation rate matrices.

A.3 Failure Mode by Model

A.4 Judge Calibration: Severity and Cross-Judge

A.5 Perturbation Verdict Summary

Type	Layer	Q-8B	Q-32B	Pool
Auth. scope	spec	28	40	34.0
Incompleteness	spec	30	42	36.0
Cond. polysemy	spec	25	30	27.5
Scopal	ling	32	35	33.5
Lexical	ling	30	33	31.5
Coreferential	ling	45	38	41.5
Spec avg	—	27.7	37.3	32.5
Ling avg	—	35.7	35.3	35.5
Layer gap	—	–8.0	+2.0	–3.0

Table 16: Per-type violation rates (%, ambiguous condition) for local models. Qwen3-8B's layer gap reverses (–8.0pp): linguistic-layer types are *more* dangerous for smaller models.

Model	Assum.	Scope	Refusal	Arb.	Perm.
GPT-5.4	45	24	8	13	10
GPT-4.1	42	23	10	15	10
Claude	22	18	28	16	16
Qwen3-32B	38	22	14	14	12
Qwen3-8B	43	20	12	15	10

Table 17: Failure mode distribution by model (%). Claude's conservative refusal rate (28%) is 2–3.5$\times$ other models'. GPT-5.4's assumption-based rate (45%) reflects aggressive pragmatic inference.

Severity	n	Prec	Rec	F1
Tier-1 critical	35	.85	.79	.82
Tier-2 moderate	40	.73	.64	.68
Tier-3 minor	25	.60	.51	.55

Table 18: Judge calibration by violation severity. Critical violations are reliably detected (F1=0.82); minor violations approach random (F1=0.55).

Agent	Judge	F1	Note
Claude S. 4.6	GPT-5.4	.75	strongest judge
GPT-5.4	GPT-4.1	.69	weaker on strong
GPT-4.1	GPT-5.4	.78	best pairing
Qwen3-8B	GPT-5.4	.71	OSS harder

Table 19: Cross-judge asymmetry. Weaker judges underperform on stronger agents' traces. Asymmetry $\leq$6pp.

Metric	Value	Threshold	Verdict
Δ_{perturb}	4.2pp	<5pp	PASS
Ratio	0.13	<0.15	PASS
CI contains 0	yes	required	PASS
CI width	12.0pp	<15pp	PASS

Table 20: Perturbation control verdict. All four criteria pass, confirming ambiguity-specific causation.